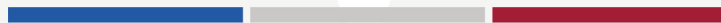




MIT SLOAN SCHOOL OF MANAGEMENT
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MODULE 1 UNIT 2
Media Set Video 1 Transcript

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Language models

JACOB ANDREAS: So before we start talking about agents, let's begin by talking about language models. Ordinarily, when we present language models in, for example, an introductory class, we talk about them as next token prediction systems. They take as input a sequence of words and they guess what word is going to come next as output.

But nowadays when most of us interact with language models, we experience them as much more complicated kinds of systems. Right? Here's an example of an interaction where I've uploaded, an image of a letterhead that I'm trying to reproduce. I've given a model some natural language instructions, and it's responded by generating, not just text as output, but also as code.

And so in the modern era, when we talk about language models, often we're thinking not just about systems that take sort of arbitrary sequences of words as input and produce text as output, but instead systems that can participate in some kind of conversation. Maybe that receive as input from a user some high-level system prompt that tells them how to behave and that generate a sequence of assistant messages in the context of some kind of chat conversation with a human user. And with language models in this sense, we can accomplish many useful tasks beyond just next word prediction.

For example, we can give them long documents as input and ask them to generate textual summaries as output. We can ask for continuations or variations on pieces of text that we're trying to write, maybe even creatively from scratch the first time. In some cases, we can ask these models questions, factual questions about the world or reasoning problems that they have to work through, and in some cases, get reliable answers as long as the information about the answers to those questions have been stored in the language model's internal weights, or even asking them to perform complicated computations like solving hard math problems.

At the same time, there are many kinds of things that we would like to do with language models that they can't do on their own. Suppose, for example, in one of these chat conversations with a language model, we want to ask it to send an email on our behalf. In the example, we issue an instruction to the language model asking it to send an email and it responds by drafting the email. If we follow-up and say, "No, I actually want you to send this email to some other person." The language model responds, "I'm sorry, I can't send emails on your behalf. The best I can do is write a well-crafted email for you to send yourself."

Similarly, if we really care about factual accuracy and we ask a question like, "What's the current headline on the Boston Globe homepage?" A typical language model will respond hopefully with something like, "Sorry, I can't provide the current headline from the Boston Globe homepage," for headlines. "Please visit this particular web address." Because the language model, as a neural network doing next token prediction, doesn't actually have any direct access to the internet on its own. Finally, anything that requires us to take real actions in the outside world, for example, if we ask a language model to take out our recycling, we should expect them not to be able to respond. And in fact, in this sort of

example here, we see that the language model says, “I’m unable to physically assist you with tasks, but I can give you some instructions for how to take out your recycling.”

So what would it take to go from the kinds of language models that we’ve been talking about here that can interact in text but can’t actually do anything in the world to language models that can actually interact with the outside, take actions on our behalf? When we think of language models in the sense that we’ve been talking about so far, we think of them as systems that take instructions from human users in and produce data, answers to our questions, or explanations as output.

But language models can also be used to take data as input and actually produce instructions as output rather than following the instructions that we give them. And as a concrete example of this, if we think back to our taking out the trash example from before, if we instead of simply instructing the language model to take out the trash, we ask it to write a computer program that when run on some sort of robot would cause the robot to take out the trash, now the language model will actually generate a reasonable first attempt at doing this. And this is a general example of a broader category of things that we can do with language models that take us the first step towards building them into language model agents.

And in the most general sense, a language model agent is a language model equipped with some kind of scaffolding that enables it to perceive and interact with the outside world where in general that external scaffolding or those actions are going to take the form of programs that the language model will write and that some external system will then run in the world in order to produce the intended side effects.

Agents

ANDREAS: It’s important to note that even though there’s a lot of excitement about language agents or language model agents in the modern era, the notion of agents as a sort of central concept in artificial intelligence goes all the way back to the earliest days of the field. If you open up even sort of classic textbooks on artificial intelligence, ones that predate anything really having to do with modern neural networks or modern large language models, often the thing that you’ll see in the very first chapter is some discussion of how to build an agent more generally.

But when we think about language model agents specifically, what we mean are just, as we said before, systems that can interact with the world where a language model forms the core sort of backbone of the decision-making process. In other words, a system where the observations and the actions that our agent takes are expressed in natural language, both on the input side and the output side.

Building an agent

ANDREAS: Now fundamentally, building a language model agent requires us to answer a few different questions, and the ways in which we answer these questions are going to determine the shape of the ultimate agent that we produce.

First, we need to know how our agent is going to perceive the world, how it takes an input from the outside in order to inform its decision-making. Second, we need to decide how the language model agent should act in the world. How it should write down its actions and

what kinds of actions it's allowed to take that will allow it to then sort of respond to its observations and produce side effects. Third, we need some way for the agent to plan and to think about the long-term consequences of its actions and reason about what those consequences are going to be before it actually begins to act. Fourth, we need to figure out some way for this agent to learn from experience so that even after it's deployed, it can continue to improve over time.

Finally, we need to think about safety, how we can architect our agent or the larger software system in which it's embedded so that we can reduce or eliminate the possibility of it taking actions with harmful side effects or give humans the opportunity to review its actions before they're taken.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.